Introduction to Economics

Class 14 — Monopolistic competition; problems on Blocks 3–4

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Plan for today

- Monopolistic competition: many sellers, each a little different
- A coworking space in the short run and the long run
- Why firms advertise
- Practice on the grid: a price taker's costs and profit
- Two curves from a table: a shift, the elasticity; a price cap

Slide 1 Title

Preview version. The full slides and handout are posted after class.

Block 4's last class. Class 12 had many sellers of an identical product and none of them able to move the price; class 13 had one seller that was the whole market. Most firms you buy from are neither: today's market has many sellers, each a little different from the others, and anyone free to open. The first half builds that case on one coworking space and ends with why such firms pay for adverts. The second half is practice: three graph problems on Blocks 3 and 4, a farm's costs and profit on a grid, two curves drawn from a table with a shift and an elasticity, and a price cap, each tried alone and then worked on the board the way you would work it on paper.

Slide 2 Plan for today

The class at a glance. Before the break, one idea built on one example: what is spent on adverts in Spain, which kind of market makes an advert worth paying for, a coworking space in the short run and after rivals open, and why it ends up with empty desks and a reason to advertise.

After the break, practice: three graph problems in the form of the practice exam on the course page, a price taker's costs and profit on a grid, two curves drawn from a table with a shift and an elasticity, and a price cap. You try each alone, then it is solved on the board the way you would solve it on paper. Problems like these are good practice for Midterm 1 on Thursday 19 November.

Advertising in Spain: how much a year?

Revealed in class

Guess before class: how many euros a year are spent on advertising in Spain, per person living here? 2.60, 26, 260 or 2,600?

Slide 3 Advertising in Spain: how much a year?

A real number about something nearly every firm does: what is spent on advertising in Spain in a year. Before class, make your guess per person living in Spain, and ask which kind of firm would pay to win one more customer: one that sells all it wants at the market price, as in class 12, or one that sets its own price, as in class 13, and why.

Revealed in class.

Monopolistic competition: many sellers, each a little different, and anyone can open one

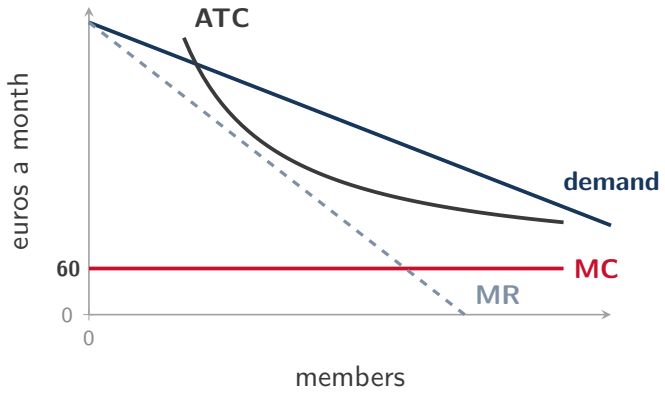
	Perfect competition (class 12)	**Monopolistic competition** (today)	**Monopoly** (class 13)
Sellers	many	many	one
Product	identical	each a little different	no close substitute
Entry	free	free	blocked
One firm's demand	flat at the price	slopes down	the whole market's

Price taker: sells all it wants at the market price (class 12, slide 6) · monopoly: the only seller, no close substitute (class 12, slide 20).

Slide 4 Monopolistic competition: many sellers, each a little different, and anyone can open one

The two outer columns are the last two classes; the middle column is today, and it borrows one feature from each. Monopolistic competition has many sellers, each selling a product that is at least a little different from the others (product differentiation: where it is, how it looks, how it is made, or simply how buyers see it), and free entry: anyone can open one or leave. The running example is coworking spaces in a city: each has its own location, light, coffee and people, and anyone can rent a floor and buy desks. Because its product differs, a space that raises its price loses some members but not all, so its own demand slopes down like a monopoly's: that is the "monopolistic" half. Because anyone can open, profits attract rivals, as under perfect competition: the "competition" half. The trap is to think that many sellers always means a price taker; that needs identical products as well.

A coworking space in the short run: how many members, at what price?



60 desks

rent, furniture and the owner's time: 3,600 a month

each member costs 60 a month

Marginal revenue (MR): what one more member adds to revenue (class 13, slide 4).

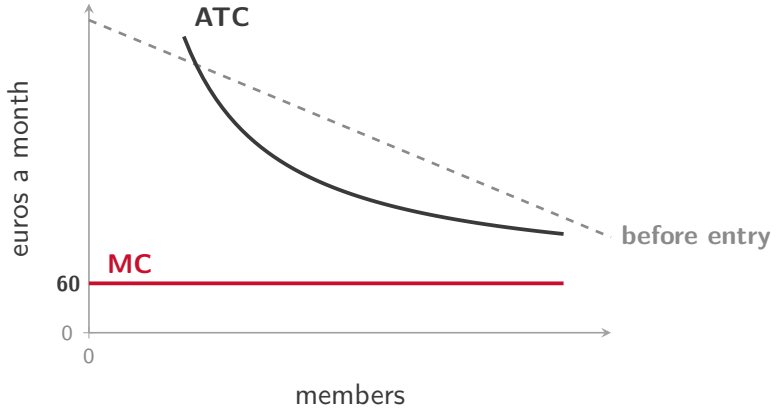
Average total cost (ATC): total cost ÷ members (class 11, slide 10) · MC: marginal cost.

Slide 5 A coworking space in the short run: how many members, at what price?

One typical space, with its numbers in the card: 60 desks, 3,600 euros a month for rent, furniture and the owner's time whatever happens, and 60 euros a month more for each member, which is its marginal cost. ATC is total cost per member, so it falls as members join. Before class, use class 13's rule for a seller whose demand slopes down: ask how many members the space wants, where its price is read from, and how its profit appears in the figure.

Revealed in class.

When new spaces open: what happens to each one's demand?



Profit here is economic profit: every cost counted, the owner's own time included (class 11, slide 6).

Slide 6 When new spaces open: what happens to each one's demand?

The profit of the short run does not stay a secret: with free entry new spaces open, and each one is a close substitute for ours. In class we draw what that does to the demand our space faces; its cost curves are unchanged. Before class, ask whether this is a move along the old demand curve or a shift of it, what the space's rule now gives, and when new spaces stop opening.

Revealed in class.

In the long run: price, cost and spare desks

Revealed in class

Compare with class 12, slide 18: where does a price taker end up in the long run?
Reading pp. 270–271.

Slide 7 In the long run: price, cost and spare desks

Once entry has stopped, the figure shows where the space ends up: its price, its cost per member and how many of its 60 desks it fills. Before class, compare with class 12, slide 18: where does a price taker end up in the long run, in price against marginal cost and in price against the lowest ATC, and ask which of those two results a seller with a downward-sloping demand can share and which it cannot. Reading pp. 270–271.

Revealed in class.

Why do firms advertise?

Revealed in class

Reading pp. 272–276: the critique and the defence of advertising, and of brand names.
What is one more customer worth to a firm whose price is above its marginal cost?

Slide 8 Why do firms advertise?

The answer to slide 3's question, from the figure of slide 7. Before class, ask what one more customer is worth to a firm whose price is above its marginal cost, and what one more buyer is worth to a price taker selling at a price equal to marginal cost. Reading pp. 272–276: the critique and the defence of advertising, and of brand names.

Revealed in class.

Monopolistic competition: many sellers, each a little different, free entry.

Short run: like a monopoly.
Long run: entry until profit is zero.

Even then the price stays above MC, and capacity is left unused.

slide 9 of 16

Slide 9 In one line

The first half in three lines, to read slowly: the first is slide 4, the second slides 5 and 6, the third slide 7. A fourth line, which answers slide 3's question, is revealed in class.

Break

Ten-minute break

slide 10 of 16

Slide 10 Break

Ten minutes. After the break: practice, three graph problems, each tried alone and then solved on the board; have paper and a pen ready.

A pomegranate farm's costs: the table — four minutes alone

A farm sells crates of pomegranates at the market price of 30 euros a crate; it is a price taker. The table gives its total cost (TC) for 0 to 5 crates a day. [5.5 points; table cells as marked]

(a) Complete the table: the marginal cost (MC) of each crate (the extra cost of that crate) and the average total cost (ATC) at 2, 3 and 4 crates.

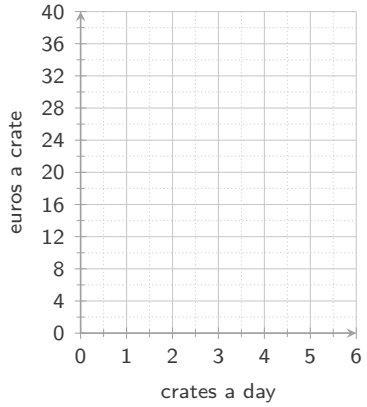
Crates a day	0	1	2	3	4	5
Total cost, TC (euros)	24	34	44	54	80	114
MC of that crate (euros) [0.5]	–					
ATC (euros a crate) [1]	–	–				–

TC: total cost · MC: marginal cost, the extra cost of one more crate · ATC: average total cost, TC ÷ crates.

The farm on the grid: MC, ATC and profit — four minutes alone

The same farm, at the market price of 30 euros a crate. Its costs (euros):

[10.5 points; drawing 4.5; boxes as marked]



crates a day	0	1	2	3	4	5
TC	24	34	44	54	80	114
MC						
ATC						

(b) Draw MC as steps: over each crate, a horizontal line at that crate's MC. Mark the ATC at 2, 3 and 4 crates with dots and join them. Draw the price as a line labelled P.

(c) Best quantity (crates a day) [2] Profit (euros a day) [2]

Shade the profit as a rectangle; write profit inside it.

(d) At 12 euros a crate: produce or shut down in the short run? Circle one; the reason in one line. [2]

(Produce) (Shut down)

MC: marginal cost · ATC: average total cost · AVC: average variable cost, variable cost ÷ crates · P: price.

Slide 11 A pomegranate farm's costs: the table — four minutes alone

The first of two slides on one farm, in the form of the practice exam on the course page: a price taker's total cost for 0 to 5 crates a day; solved in class on the board. Try it alone before class: fill the marginal-cost row, the extra cost of each crate, and the average total cost where the table asks for it; then ask what the cost at zero crates tells you.

Revealed in class.

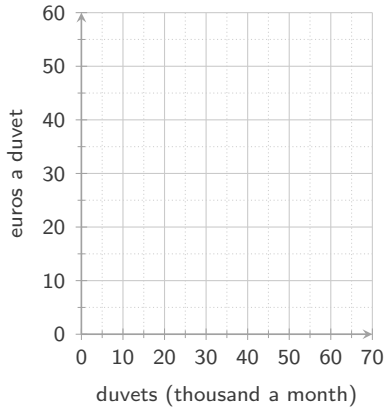
Slide 12 The farm on the grid: MC, ATC and profit — four minutes alone

The same farm on a grid, at the market price of 30 euros a crate; solved in class on the board. Fill the previous slide's cells first. Then try it alone before class on a sketch of the grid: start with the scale on each axis, draw the marginal cost as steps, the average total cost as joined dots and the price as a line, each with its label; find the best quantity and the profit, and decide what the farm does when the price falls, with the reason in one line.

Revealed in class.

Duvets: two curves from a table, then a shift — six minutes alone

The market for duvets (quantities in thousand duvets a month). The table gives the quantities demanded and supplied at three prices; both curves are straight lines.



[12 points; drawing 6; boxes as marked]

Price (euros a duvet)	25	35	45
Quantity demanded (thousand a month)	60	40	20
Quantity supplied (thousand a month)	15	25	35

(a) Draw the demand curve and the supply curve through the table's points, label them D and S, and mark the equilibrium E_1.

(b) The price of polyester, an input used to make duvets, falls. At every price, sellers now offer 30 thousand more duvets a month than before. Draw the new supply curve, label it S_2, and mark the new equilibrium E_2.

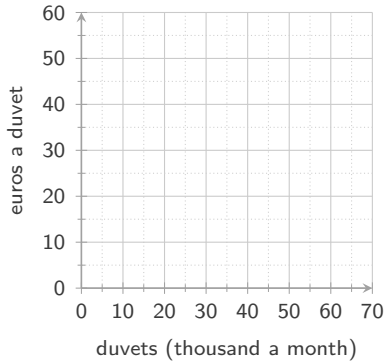
	Price (euros) [1.5]	Quantity (thousand) [1.5]
(a) Before: E_1		
(b) After: E_2		

Slide 13 Duvets: two curves from a table, then a shift — six minutes alone

A market in the practice exam's form: a table of the quantities demanded and supplied at three prices, then an event that changes what sellers offer; solved in class on the board. Try it alone before class on a sketch of the grid: start with the scale on each axis, plot the table's points with the quantity across and the price up, draw and label both lines and mark where they cross; then draw the new supply curve and the new crossing, and ask which curve moved and which did not.

Revealed in class.

The duvets: the elasticity from E_1 to E_2 — two minutes alone



[2 points; working shown]

(c) Demand did not move, so E_1 and E_2 are two points on the demand curve. Compute the price elasticity of demand between them by the midpoint method, and write it in the box as a positive number (a fraction or a decimal).

Elasticity (a positive number) [2]

Working:

E_1, E_2: the equilibrium before and after · price elasticity of demand: midpoint method, as a positive number.

Slide 14 The duvets: the elasticity from E1 to E2 — two minutes alone

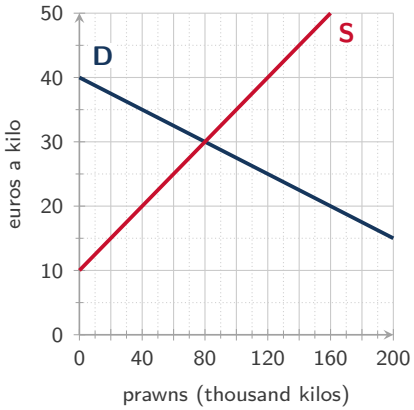
The same market: the two crossings of the previous slide are two points on one demand curve, and the slide asks for the price elasticity of demand between them by the midpoint method. Try it alone before class with your own two points: the change in quantity over the average quantity, the change in price over the average price, then one divided by the other; decide which goes on top, and write each line of working.

Revealed in class.

Prawns at Christmas: a price cap — five minutes alone

The grid shows a city's market for prawns in Christmas week (quantities in thousand kilos): demand D and supply S. The city sets a maximum price (a price cap) of 20 euros a kilo.

[9 points; drawing 2; boxes as marked]



(a) Draw the cap as a line and label it cap.

(b) The quantity supplied, the quantity demanded and the shortage at the cap (thousand kilos).

Supplied [1.5] Demanded [1.5] Shortage [1]

(c) Only the quantity supplied is sold, and it goes to the buyers who value it most. The deadweight loss (DWL) with the cap, in thousand euros; write DWL inside its area.

DWL [3]

Thousand kilos × euros a kilo = thousand euros. A triangle is ½ × base × height.

Slide 15 Prawns at Christmas: a price cap — five minutes alone

A third market in the practice exam's form: the grid prints the demand and the supply for prawns in Christmas week, and the city sets a maximum price; solved in class on the board. Try it alone before class: start with the scale on each axis (it differs from the duvets' grid), draw the cap as a line with its label, read the two quantities at the cap from the right curves, and find the triangle of trades that no longer happen before you compute it.

Revealed in class.

Before Monday 16 November

- Monday 16 November: review for Midterm 1 (Blocks 1–4), the full two hours
- Thursday 19 November: Midterm 1 (Blocks 1–4), in class
- Practise in Sextant: the units the course page lists

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Slide 16 Before Monday 16 November

Monday 16 November: the review session for Midterm 1, Blocks 1 to 4, the whole class. Thursday 19 November: Midterm 1, Blocks 1 to 4, in class. Problems like today's three are good practice: the practice exam on the course page has the same kind of pages, so do it in one sitting. Sextant has more, in the units the course page lists.

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